

S1.1.

P2. $y = 3e^{-2x}$

$$y' = -6e^{-2x}$$

$$\Rightarrow y' + 2y = -6e^{-2x} + 6e^{-2x} = 0$$

P3. $y_1 = \cos 2x$

$$y_1' = -2\sin 2x, y_1'' = -4\cos 2x$$

$$\Rightarrow y_1'' + 4y_1 = -4\cos 2x + 4\cos 2x = 0$$

$$y_2 = \sin 2x$$

$$y_2' = 2\cos 2x, y_2'' = -4\sin 2x$$

$$\Rightarrow y_2'' + 4y_2 = -4\sin 2x + 4\sin 2x = 0$$

P7. $y_1 = e^x \cos x$

$$y_1' = -e^x \sin x + e^x \cos x$$

$$y_1'' = -e^x \sin x - e^x \cos x + e^x \cos x - e^x \sin x = -2e^x \sin x$$

$$\Rightarrow y_1'' - 2y_1' + 2y_1 = -2e^x \sin x - 2(-e^x \sin x + e^x \cos x) + 2e^x \cos x = 0$$

$$y_2 = e^x \sin x$$

$$y_2' = e^x \sin x + e^x \cos x$$

$$y_2'' = e^x \sin x + e^x \cos x + e^x \cos x - e^x \sin x = 2e^x \cos x$$

$$\Rightarrow y_2'' - 2y_2' + 2y_2 = 2e^x \cos x - 2(e^x \sin x + e^x \cos x) + 2e^x \sin x = 0$$

P10. $y_1 = x - \ln x$

$$y_1' = 1 - \frac{1}{x}, y_1'' = \frac{1}{x^2}$$

$$\Rightarrow x^2 y_1'' + x y_1' - y_1 = x^2 \left(\frac{1}{x^2}\right) + x \left(1 - \frac{1}{x}\right) - (x - \ln x) = \ln x$$

$$y_2 = \frac{1}{x} - \ln x$$

$$y_2' = -\frac{1}{x^2} - \frac{1}{x}, y_2'' = \frac{2}{x^3} + \frac{1}{x^2}$$

$$x^2 y_2'' + x y_2' - y_2 = x^2 \left(\frac{2}{x^3} + \frac{1}{x^2}\right) + x \left(-\frac{1}{x^2} - \frac{1}{x}\right) - \left(\frac{1}{x} - \ln x\right) = \ln x$$

P17. Solution:

$$y(x) = Ce^{-x}, y'(x) = -Ce^{-x}$$

$$y' + y = -Ce^{-x} + Ce^{-x} = 0$$

$$y(0) = 2 \Rightarrow C \cdot e^0 = 2 \Rightarrow C = 2$$

$$\Rightarrow y(x) = 2e^{-x}$$

is the solution of the initial value problem.

P22. Solution:

$$y(x) = \ln(x+c), y'(x) = \frac{1}{x+c}$$

$$e^y y' = e^{\ln(x+c)} \cdot \frac{1}{x+c} = 1$$

$$y(0) = 0 \Rightarrow \ln(0+c) = 1 \Rightarrow c = 1$$

$$\Rightarrow y(x) = \ln(x+1)$$

is the solution of the initial value problem.

P27. Solution:

$$\frac{dy}{dx} = x + y$$

P28. Solution:

$$\frac{dy}{dx} = \frac{y-0}{x-x/2} = \frac{y}{x/2} = \frac{2y}{x}$$

P32. Solution:

$$\frac{dp}{dt} = k p^2$$

where k is a constant.

P43. Solution:

$$a) X(t) = \frac{1}{c-kt}$$

$$X' = \frac{k}{(c-kt)^2} = kX^2$$

$$b) X(0) = 0$$

$$\Rightarrow \frac{1}{c-kt} = 0 \Rightarrow \frac{1}{c} = 0$$

$$c = \infty$$

$$\Rightarrow X \equiv 0$$

S1.2

P1. Solution:

$$y(x) = \int 2x+1 dx + C$$

$$= x^2 + x + C$$

$$y(0) = 3 \Rightarrow C = 3$$

$$\Rightarrow y(x) = x^2 + x + 3$$

P2. Solution:

$$y(x) = \int (x-2)^2 dx + C$$

$$= \frac{1}{3}(x-2)^3 + C$$

$$y(2) = 1 \Rightarrow C = 1$$

$$\Rightarrow y(x) = \frac{1}{3}(x-2)^3 + 1$$

P5. Solution:

$$y(x) = \int \frac{1}{\sqrt{x+2}} dx = 2\sqrt{x+2} + C$$

$$y(2) = -1 \Rightarrow 2 \cdot \sqrt{4} + C = -1$$

$$\Rightarrow C = -5$$

$$\Rightarrow y(x) = 2\sqrt{x+2} - 5$$

$$P8. \text{Solution: } y(x) = \int \cos 2x dx + C$$

$$= \frac{1}{2} \sin 2x + C$$

$$y(0) = 1 \Rightarrow C = 1$$

$$\Rightarrow y(x) = \frac{1}{2} \sin 2x + 1$$

$$P10. \text{Solution: } y(x) = \int x e^{-x} dx + C$$

$$= x(-e^{-x}) - \int (-e^{-x}) dx + C$$

$$= -x e^{-x} - e^{-x} + C$$

$$y(0) = 1 \Rightarrow -1 + C = 1 \Rightarrow C = 2$$

$$\Rightarrow y(x) = e^{-x}(-x-1) + 2$$

$$P11. \text{Solution: } a(t) = 50, v_0 = 10, x_0 = 20$$

$$x(t) = \frac{1}{2} a t^2 + v_0 t + x_0$$

$$= 25 t^2 + 10 t + 20$$

$$P13. \text{Solution: } a(t) = 3t; v_0 = -15, x_0 = 5$$

$$v(t) = \int_0^t a(s) ds + v_0 = \int_0^t 3s ds + (-15)$$

$$= \frac{3}{2} s^2 \Big|_{s=0}^{s=t} - 15 = \frac{3}{2} t^2 - 15$$

$$x(t) = \int_0^t v(s) ds + x_0 = \int_0^t \left(\frac{3}{2} s^2 - 15 \right) ds + 5$$

$$= \left(\frac{1}{2} s^3 - 15s \right) \Big|_{s=0}^{s=t} + 5$$

$$= \frac{1}{2} t^3 - 15t + 5$$

$$P16. \text{Solution: } a(t) = \frac{1}{\sqrt{t+4}}, v_0 = -1, x_0 = 1$$

$$v(t) = \int a(t) dt + C_1 = \int \frac{1}{\sqrt{t+4}} dt + C_1$$

$$= 2\sqrt{t+4} + C_1$$

$$v(0) = -1 \Rightarrow 4 + C_1 = -1 \Rightarrow C_1 = -5$$

$$\Rightarrow v(t) = 2\sqrt{t+4} - 5$$

$$x(t) = \int v(t) dt + C_2 = \int 2\sqrt{t+4} - 5 dt + C_2$$

$$= \frac{4}{3} (t+4)^{3/2} - 5t + C_2$$

$$x(0) = 1 \Rightarrow \frac{4}{3} \cdot 8 + C_2 = 1 \Rightarrow C_2 = -\frac{29}{3}$$

$$\Rightarrow x(t) = \frac{4}{3} (t+4)^{3/2} - 5t - \frac{29}{3}$$

$$P25. \text{Solution: } x(t) = \frac{1}{2} a t^2 + v_0 t + x_0, v(t) = at + v_0$$

$$\text{Notice } x_0 = x(0) = 0, v_0 = v(0) = 1000000 \text{ m/h}$$

$$a = -10 \text{ m/s}^2 = -250/9 \text{ m/s}^2$$

$$\Rightarrow x(t) = -5t^2 + \frac{250}{9} t$$

$$v(t) = -10t + \frac{250}{9}$$

$$\text{To reach a stop, we have } v(t) = 0$$

$$\Rightarrow -10t + \frac{250}{9} = 0 \Rightarrow t = \frac{25}{9}$$

$$\Rightarrow x\left(\frac{25}{9}\right) = -5 \cdot \left(\frac{25}{9}\right)^2 + \frac{250}{9} \cdot \frac{25}{9} \approx 38.58 \text{ m}$$

P28. solution:

$$V_0 = -40 \text{ ft/s}, g = 32 \text{ ft/s}^2, X_0 = 555 \text{ ft}$$

$$X(t) = -\frac{1}{2}gt^2 + V_0t + X_0$$

$$= -16t^2 - 40t + 555$$

The baseball reaches the ground, i.e.

$$X(t) = 0$$

$$\Rightarrow -16t^2 - 40t + 555 = 0$$

$$t = \frac{40 \pm \sqrt{40^2 - 4 \times (-16) \times 555}}{2 \cdot (-16)}$$

$$\Rightarrow t_1 \approx -7.27 \text{ s}, \boxed{t_2 \approx 4.77 \text{ s}}$$

The time has to be positive, thus
it takes about 4.77 s.

$$V(t) = -gt + V_0$$

$$= -32t - 40$$

$$V(4.77) = (-32)(4.77) - 40$$

$$\approx -192.64 \text{ ft/s}$$